# Server Hardening

### Objectives:

At the end of this episode, I will be able to:

1. Identify server hardening techniques.

Exam Objective: *3.5 - Given a scenario, apply server hardening methods.*

Description: In this episode, the viewer will identify a collection of techniques called server hardening to increase resiliency to threats, vulnerabilities and attacks such as operating system hardening, application hardening, and hardware hardening.

---

- OS hardening
  - Install only required software
    - Pause Ubuntu installation (mention small footprint)
    - Show terminal (no GUI, less software)
  - Apply driver updates (Show Windows update)
    - Device manager to view current device state
    - Windows updates for immediate updates
    - Third party vendors for specilized drivers
  - Apply OS updates (Windows Update, or WSUS)
  - Firewall configuration (Windows Defender Firewall with Adv. Security)
    - Configure needed ports
    - Close unneeded ports
  - Disable unused services
    - Services console
  - Close unneeded ports
    - Uninstall the service or disable the service
- Application hardening
  - Install latest patches
  - Disable unneeded services, roles, or features
- Hardware hardening
  - Disable unneeded hardware
  - Disable unneeded physical ports, devices, or functions
    - Device manager (locate Class and Class GUID)
    - Vendor documentation : [https://docs.microsoft.com/en-us/windows-hardware/drivers/install/system-defined-device-setup-classes-available-to-vendors](https://docs.microsoft.com/en-us/windows-hardware/drivers/install/system-defined-device-setup-classes-available-to-vendors)
      - Group policy

```
 Computer Config/Policies/Admin Templates/Removable Storage Device Access
```

```
Computer Configuration/Administrative Templates/System/Device Installation/Device Installation Restrictions/ allow installation of devices that match any of these Device IDs
```

    - BIOS/UEFI settings
        - Set BIOS/UEFI password
        - Set boot order
        - Enable Secure Boot
        - TPM

---

### External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Netwrix Server 2019 Hardening Guide : https://www.netwrix.com/windows_server_hardening_checklist.html
- Ubuntu Server Hardening Guide: https://gist.github.com/lokhman/cc716d2e2d373dd696b2d9264c0287a3